Chapter 1

Preoperative assessment

Cardiovascular history *2*
Systems enquiry *4*
Examination *6*
Routine laboratory tests *8*
Plain chest radiograph *12*
Computed tomography *16*
Cardiac catheterization *18*
Interpreting coronary imaging *20*
Cardiac hemodynamics *22*
Echocardiography *24*
Echo interpretation *26*
Perfusion and viability *28*
General pre-op checklist *30*
EKG interpretation is described on 🕮 p210. Consent is discussed on 🕮 p257.

Cardiovascular history

There are three main aims of obtaining a cardiovascular history:
- Establish the indications for surgery.
- Identify issues that need management outside standard protocols.
- Define preoperative estimate of risk (mortality, stroke, etc.).

Angina

Angina is frequently not the classic 'central, crushing, chest pain radiating to the jaw and down the left arm'. It may present as heaviness, indigestion, isolated arm, neck or jaw discomfort, nausea, or fatigue. Ask about precipitants such as exertion, stress, emotion, and cold air, as well as relieving factors such as rest and sublingual nitroglycerine (NTG). How many times a day is NTG spray used? Angina is graded (see Box 1.1). ***Class III and IV (heart failure) symptoms*** are associated with increased risk.

Breathlessness

Ask about exertional dyspnea and classify it (see Box 1.1). ***Heart failure is defined as class III or IV symptoms***. Ask about other precipitants: paroxysmal nocturnal dyspnea, orthopnea, palpitations, chest infections, chronic obstructive pulmonary disease (COPD), pulmonary embolism (PE), medication such as β-blockers. Ask about associated symptoms such as peripheral edema, palpitations, wheeze, and sputum. Ask about relieving factors such as NTG, rest, sitting up.

Previous myocardial infarction

Document the number, dates, and anatomy of any previous myocardial infarctions (MIs). Check chart and EKG (Q waves). Risk of surgery is higher after recent ST segment elevation MI (STEMI) than non-ST segment elevation MI (STEMI), and highest early after MI, falling after 1 week (📖 p297).

Previous chest surgery

Ask, check chart, and look carefully at ***whole chest and back*** for scars. Previous cardiac surgery is associated with incremental operative mortality and requires several changes to standard operative approach (📖 p342). Most patients should undergo CT (computed tomography) chest to evaluate relationship of aorta and right ventricle (RV) to sternum, and location of patient bypass grafts (requires iv contrast). Obtain operative report.

Box 1.1 Functional classifications

Canadian Cardiovascular Society (CCS) applies to angina only
- Class I: no angina with ordinary activity.
- Class II: slight limitation of ordinary activity.
- Class III: marked limitation of ordinary activity.
- Class IV: angina on minimal exertion or at rest.

New York Heart Association (NYHA) dyspnea and/or angina
- Class I: no limitation during ordinary activity (asymptomatic).
- Class II: slight limitation during ordinary activity.
- Class III: marked limitation of ordinary activity.
- Class IV: no physical activity without symptoms.

Previous percutaneous interventions

Note left ventricle (LV) function and coronary anatomy from the angiography report. Document the number, dates, type, and result of previous percutaneous interventions (angioplasty or stenting). Emergency surgery after failed percutaneous coronary intervention (PCI) is high risk. Recent PCI usually mandates at least 1 month of clopidogrel (📖 p18), and can impact conduct of coronary bypass surgery (📖 p651).

Syncope and dizziness

'Black-outs' may be a result of epilepsy, transient ischemic attacks (TIAs), hypoglycemia, vasovagal (standing, emotion, coughing, micturition) syndromes. True ***cardiac syncope*** from aortic stenosis (AS) is usually exertional. Syncope when the patient is sitting or lying suggests arrhythmias. Dizziness may be associated with syncope (presyncopal episodes), but is fairly non-specific. Common causes are antianginals, antihypertensives, and diuretics causing postural hypotension. Dizziness may reflect low cardiac output states (failure, arrhythmias, ischemia) or cerebrovascular disease. Diabetic autonomic neuropathy is probably an under-recognized cause.

Palpitations and atrial fibrillation

To the patient palpitations may be skipped beats, tachycardia, irregular rhythm, or awareness of their heart particularly at night. Ask about onset: rapid onset and offset suggests paroxysmal arrhythmias, gradual onset and offset suggests sinus tachycardia. Preoperative atrial fibrillation (AF) may require ablation and/or left atrial (LA) appendage ligation, and warfarin (📖 pp262, 484).

Peripheral edema

This may be associated with changes in diuretic therapy, weight gain, and exacerbations of dyspnea, and helps quantify the heart failure.

Extracardiac arteriopathy

Ask specifically about TIAs, strokes, claudication, and vascular interventions. A history of TIAs warrants a carotid duplex to exclude carotid disease which may be an indication for carotid intervention (📖 p 279). Extracardiac arteriopathy is important for risk scoring (📖 p359), affects the risk of stroke and limb ischemia quoted when consenting the patient, and influences the choice of conduit (📖 p302) and cannulation site (📖 p120).

Coronary risk factors

Reversible risk factors that should be addressed in hospital include: smoking, hypercholesterolemia, obesity, hypertension, and diabetes. Irreversible risk factors include male sex, age, and family history. Be precise: 'non-smoker' is not the same as 'smoked 60 a day, gave up 2 weeks ago.'

Valve patients

Note the type and degree of valve disease, and LV function from the echo. Ask about rheumatic disease and endocarditis. Check that patients have been recently cleared by a dentist (edentulous patients to do not need to be seen by a dentist), check that they have no active sources of infection, they have had an angiogram if they are males >45 years, postmenopausal females, have angina type symptoms or multiple coronary risk factors, and that the ***type of valve replacement*** (📖 p370) has been discussed.

Systems enquiry

The main aims of the systems enquiry are:
- Identifying problems that alter standard perioperative management.
- Identifying contraindications to elective surgery.
- Identifying problems that could potentially delay discharge.

Previous medical history

Ask about any other medical problems, and repeat the question by asking if the patient has ever been in hospital with any problems other than their heart, and whether they have had any operations, ***ever***. Be persistent as patients often forget important chronic illnesses and operations.
- Preoperative management of ***diabetic patients*** is described on 📖 p31.
- Previous ***chest radiotherapy*** mandates CT scan: aorta may be calcified.
- ***Malignant disease*** should be carefully evaluated: in most cases metastatic spread is a relative contraindication to cardiac surgery.
- ***Thyroid disease*** may dictate the choice of antiarrhythmic (📖 p205).
- Patients with ***hip replacements*** should be transferred with a pillow between their legs: adducting the hip may result in dislocation.
- Patients with ***blood dyscrasias*** (thrombocytopenia, factor deficiencies, thrombophilias) should be reviewed by a hematologist.

Medication and allergies

Take full details of all drugs, doses, and regimens as well as allergies to drugs, iodine, contrast, and fish (protamine is made from salmon sperm).
- ***Latex allergies*** require changes to equipment: warn operating room.
- History of ***heparin-induced thrombocytopenia*** may require non-heparin (bivalirudin or argatroban) bypass, which mandates careful preplanning.
- Continue ***antianginals*** until surgery, and stop postoperatively.
- Continue β-***blockers*** until surgery, and reintroduce as blood pressure (BP) and rhythm permit postoperatively.
- Stop ***ACE inhibitors*** 24–48 hours before surgery to avoid profound peripheral dilatation in the immediate postoperative period.
- ***Aspirin*** and ***clopidogrel*** are discontinued by many surgeons 5–10 days preoperatively as they predispose to increased postoperative bleeding, but there is a slightly higher risk of acute ischemic events and in high-risk patients (e.g. recent PCI, tight left main stem (LMS)) they may be continued.
- ***Warfarin*** should be stopped 5 days before surgery (📖 p31). In patients at higher risk of thromboembolic event (e.g. prior TIA, deep vein thrombosis (DVT)) admission 24–48 hours pre-op may be required for iv heparin.
- ***Steroids*** should be reduced where possible preoperatively: patients taking >5mg of prednisone daily should be prescribed 48 hours 50–100mg hydrocortisone iv 6-hourly, or an equivalent dose continuous infusion, starting with the premedication to avoid Addisonian crisis as a result of adrenal suppression.
- Withhold long-acting ***hypoglycemics*** on the morning of surgery and commence an insulin sliding scale where appropriate (📖 p31).

Social history

- ***Jehovah's witness*** patients do not accept blood or blood products, and some may not agree to use of cell salvage: these patients must be identified preoperatively, counselled, preferences clearly documented, and every effort made to ensure preoperative hamatocrit >40%. Several weeks of erythropoietin therapy may be indicated.
- Heavy/binge drinkers are at risk of alchohol withdrawal. Haloperidol lowers the seizure threshold and prolongs the QT-prescribe diazepam 2mg po 8-hourly, thiamine and folate.
- If the patient *lives alone*, plan for rehabilitation. If the patient mobilizes with a walking stick or frame contact physiotherapists early, as sternal union may be compromised even if mobility is not.

Respiratory history

An *active upper respiratory tract infection* (URTI) is sufficient reason to cancel the operation, so ask about cough, colds, and sputum. If you suspect a URTI measure temperature, C-reactive protein (CRP), and white cell count (WCC), and get a chest X-ray (CXR). Assess ***COPD*** by asking about hospital admissions, inhalers, nebulizers, and home O_2. If dyspnea is the prominent symptom, and the patient has COPD get lung function tests (📖 p652) including blood gases.

Gastrointestinal history

Cardiac surgery patients are at risk of upper gastrointestinal (GI) bleeds from *peptic ulcer disease*, and prolonged ileus, as well as acute deterioration in pre-existing disease, e.g. pancreatitis, and cholecystitis, so ask specifically about GI problems, particularly dyspepsia, ulcers, and pancreatitis. Start all patients on prophylactic proton pump inhibitors (PPIs), and make a note on the drug chart to avoid prescribing non-steroidal anti-inflammatory drug (NSAID) analgesics in at-risk patients. Aspirin should be enteric coated and given with food in these patients.

Renal history

Check the notes for evidence of ***renal dysfunction***. If there has been an acute deterioration it may be worth delaying the patient to optimize this by stopping nephrotoxic drugs including angiotensin-converting enzyme (ACE) inhibitors and NSAID analgesia, starting iv fluids with careful fluid balance, and in some cases dialysis. ***Dialysis patients*** should be dialysed the morning of surgery or the night before. Patients undergoing preoperative cardiac catheterization are at particular risk of temporary impairments in renal function because of the combination of hypovolemia due to preprocedure fasting, and nephrotoxic contrast. Prescribe iv fluid in patients with borderline renal function, and consider iv acetylcysteine prior to contrast studies. Patients with a history suggestive of ***prostatic enlargement*** should be warned that they may need a suprapubic catheter.

Neurological history

In addition to identifying TIAs, strokes, and residual deficit identify dementia, which in advanced cases is a contraindication to cardiac surgery.

Examination

The main purpose of the examination is:

- To identify disease that needs further investigation and management.
- To help plan surgery.
- To identify problems that may delay recovery.

For the average elective patient a shortened version of the full cardiovascular examination described is sufficient (see Box 1.2).

Box 1.2 Short examination

- Assess pulse rate and rhythm.
- Document BP.
- Assess jugular venous pressure with the patient at 45°.
- Auscultate heart sounds, carotids and breath sounds. Look for scars.
- Examine the abdomen (tenderness, hernias, organomegaly, scars).
- Feel peripheral pulses including the abdominal aorta.
- Asses pedal edema.
- Assess conduits.

General inspection

From the end of the bed assess whether the patient is comfortable at rest, requiring oxygen, well or moribund, obese or cachexic, or has features suggestive of Down's syndrome (atrial or ventricular septal defect (ASD/VSD), or Marfan syndrome (aortic aneurysms, dissection, and incompetence). Position the patient at 45° in the bed.

Hands and forearms

Look for nicotine staining, stigmata of endocarditis, e.g. splinter hemorrhages, Osler's nodes (tender lumps in the fingertip pulps), Janeway lesions (red macules on the palms or flexor aspects of the wrist), tendon xanthoma. Petechiae may be present in infective endocarditis.

Pulse

Note rate and rhythm at the radial pulse, and character and volume at the brachial or carotid pulse. AS classically has a slow rising, thready pulse. Aortic incompetence has a large volume bounding pulse. Feel the peripheral pulses including the abdominal aorta.

Blood pressure

Note uncontrolled hypertension: systolic >180mmHg should be treated with nifedipine 10mg s/l preoperatively. A wide pulse pressure is suggestive of aortic incompetence. Use a broad cuff for fat arms as BP will otherwise be underestimated. Check BP in both arms if aortic dissection or coarctation is suspected.

Face

Malar flush (cyanosis in a butterfly rash distribution) is characteristic of mitral stenosis. Look at the eyes: jaundice may be caused by hemolysis from prosthetic heart valves; xanthelasma and corneal arcus are common signs of hyperlipidemia, Roth spots may be seen in infective endocarditis, and rarely Argyll Robertson pupils are a feature of syphilitic aortic

incompetence. Look at the mouth: check for central cyanosis by examining the tongue and gums, look for the high arched palate of Marfan and check for poor dentition particularly in valve patients.

Neck

Assess the jugular venous pulse with the patient at 45°.

Chest

Inspect the chest for scars (left and right lateral thoracotomy and median sternotomy), implantable cardioverter defibrillator (ICD) and pacemaker insertion, pectus, visible pulsations. Expose the patient adequately or you will miss these.

Palpate the apex beat, which is the point furthest from the manubrium where the heart can be felt, normally 5th intercostal space in the mid-clavicular line. This is shifted laterally in cardiomegaly or mediastinal shift. AS and hypertensive cardiomyopathy results in a sustained, forceful apex beat. A dilated LV (e.g. aortic regurgitation (AR)) normally has a downwards displaced apex beat. In mitral stenosis (MS) the apex beat is described as tapping. Left parasternal heave may be felt in right ventricular hypertrophy. Thrills are palpable murmurs.

Ascultate Timing heart sounds against the radial pulse, use the bell to listen to low-pitched sounds and the diaphragm for high-pitched sounds. Listen over the apex (diaphragm and bell), upper and lower left sternal borders, under the clavicles, over the carotids, and in the axilla (diaphragm ± bell), and then listen again at the apex with the patient in the left lateral position, and at the lower left sternal border with the patient sitting forward in full expiration. Murmurs can be graded on a scale of 1–6: grade 1 is just audible in a quiet room with the patient breath holding, grade 3 is easy to hear without a thrill, grade 6 is audible without a stethoscope. Murmurs and carotid bruits need ultrasound (US) imaging.

Heart murmurs (📖 pp373, 375, 398, 401, 416, 559)

- ***Ejection systolic:*** AS (loudest in aortic area in expiration, radiates to carotids), aortic sclerosis, pulmonary stenosis, ASD, 'innocent' flow murmur (anemia, hyperthyroid, pregnancy).
- ***Pansystolic:*** mitral regurgitation (MR) (loudest at apex, radiates to axilla), tricuspid regurgitation (TR), VSD, patent ductus arteriosus (PDA).
- ***Early diastolic:*** AR, pulmonary regurgitation.
- *Mid diastolic:* MS, tricuspid stenosis (TS), Austin Flint murmur of AR.

Conduit

Stand the patient up so that you can see both legs from ankle to groin. Look for varicosities and decide which distribution they are in (long or short saphenous). To test the suitability of the radial artery perform Allen's test by asking the patient to clench their fist tightly, occluding both radial and ulnar pulses, asking the patient to open their hand and noting how long it takes for the palm of the hand to pink up once the ulnar occlusion is released. Refill time <5s is suitable for harvest.

General examination

Listen to the chest and palpate the abdomen.

Routine laboratory tests

Routine preoperative investigations

- Complete blood count (CBC).
- Coagulation screen (activated partial thromboplastin time (APPT) and prothrombin time (PT)). International normalized ratio (INR) if taking warfarin.
- Biochemistry (Chem 7: Na^+, K^+, Cl^-, HCO_3, blood urea nitrogen (BUN), Cr, glucose).
- Liver function tests (LFTs).
- Cross match blood (XM) 2 units (4 units + fresh frozen plasma (FFP)/ platelets (plts) in reoperations or thoracoabdominals).
- Methicillin-resistant *Staphylococcus aureus* (MRSA) screening swab, urinalysis, pregnancy, and sickle cell testing.

Management of common biochemical derangements

See Table 1.1.

Table 1.1 Management of common biochemical derangements

Result	Causes	Management
↑Na >145mmol/L	Commonly diarrhea, vomiting. Less commonly diabetes insipidus.	Oral water or iv 5% dextrose. Discuss with anesthesiologist
↓Na <130mmol/L	Commonly diuretics, diarrhea, vomiting Less commonly nephritic syndromes (congestive cardiac failure (CCF), cirrhosis, renal failure), Addison's, syndrome of inappropriate antidiuretic hormone secretion (SIADH) (common postoperatively)	Assess renal function (📖 p234) and treat accordingly. Fluid restrict if appropriate (📖 p271). Discuss with anasthesiologist
↑K >5mmol/L	Commonly K-sparing diuretics and ACE inhibitors, excess K therapy, artifact (hemolysis in tube) Less commonly renal failure, diabetic acidosis, Addison's disease	See 📖 p271. It should be possible to correct hyperkalemia over 12h
↓K <3.5mmol/L	Commonly diuretics, vomiting, diarrhea Less commonly Conn's, Cushing's syndromes, renal tubular failure	See 📖 p270. It should be possible to correct hypokalemia over 12h

Table 1.1 (*Contd.*)

↑Cr >1.2mg/dL BUN >21mg/dL	Renal impairment. Disproportionately ↑U, suggests dehydration as primary cause. Isolated ↑U occurs with thiazide diuretics, gout, lymphoma, leukemia	See 🕮 p270.
↑LFTs or bilirubin, or gamma GT	Commonly drugs (paracetamol, OCP, co-amoxiclav, chlorpromazine, erythromycin), hemolysis, viral hepatitis, gallstones, Gilbert's syndrome, sepsis, EtOH Less commonly malignancy causing biliary tract obstruction,	Ask about iv drug use, transfusions, travel, EtOH. Examine abdomen for ↑spleen/liver, signs of chronic liver disease, test urine for bilirubin and urobilinogen. Request liver US. Consider serology.

The decision to cancel surgery in order to investigate or treat abnormalities thrown up in preoperative screening is based on judging the risk of delaying surgery against the increased risk of surgery. The symptomatic, unstable patient with a tight LMS lesion is less likely to be cancelled for a mild neutrophilia than the elective valve patient, for example. Discuss any move to cancel with the surgical team.

Management of common hematological derangements

See Table 1.2.

Table 1.2 Management of common hematological derangements

Result	Causes	Management
↓Hb <9.5g/dL	Mean corpuscular volume (MCV) <75: Fe deficiency, thalassemia MCV 75–95: chronic disease, renal failure, bone marrow failure, hemolysis, hypothyroidism, dilutional (acute) MCV >95: ↓B_{12} or folate, EtOH, liver disease, hypothyroidism, myelodysplasias, bone marrow infiltration	Undiagnosed Fe deficiency should be fully investigated as GI malignancy is a common cause in this age group Treat the causes where possible, and avoid transfusion unless Hb ⇊. The perfusionist may want extra blood to prime the pump

(*Continued*)

Table 1.2 (*Contd.*)

↑Hb >14g/dL	Relative (↓plasma volume): dehydration (EtOH and diuretics), Absolute (↑red blood cell (RBC) mass): primary polycythemia rubra vera, secondary to smoking, COPD, altitude, tumours	PCV patients may need formal anticoagulation postoperatively to prevent thromboembolic events
↑WCC >12x10^9/L	Neutrophilia: bacterial sepsis, trauma, surgery, burns, inflammation, MI, polymyalgia, steroids, hemorrhage Lymphophilia: viral infection (Epstein–Barr virus (EBV), cytomegalovirus (CMV)), chronic lymphocytic leukemia (CLL) Eosinophilia: asthma, allergy, parasitic infections, eczema, malignancy	Recheck history and examination for evidence of sepsis, take temperature, look at CXR and urine dipsticks. Elective valve patients: check with team: may cancel. If due to a URTI, start ABCs, and recheck WCC in 2–3 weeks
↓WCC<2 x 10^9/L	Rarely neutropenic sepsis, steroids, sulfonamides, systemic lupus erythematosus (SLE)	Discuss with hematologist—emergency in sepsis
↑Plts >400x10^9/L	Essential thrombocythemia, sepsis, inflammatory disease (e.g. rheumatoid arthritis). 50% have malignancy	Discuss with hematologist
↓Plts <120x10^9/L	Heparin-induced thrombocytopenia (📖 p288), sepsis and DIC (📖 p288), drugs, SLE, lymphoma, bone marrow failure.	Discontinue heparin and request heparin-induced thrombocytopenia (HIT) screen. In septic patients send fibrin degradation products. Request plts
Clotting	↑PT = warfarin, ↓vit K, liver disease ↑APTT = heparin, ↓factor VIII, IX ↑INR = warfarin, liver disease	Investigate as for liver disease, measure clotting factors. Talk to hematologist
XM	Atypical antibodies may mean more time and further samples are required to obtain blood that gives a satisfactory cross-match.	

This page intentionally left blank

Plain chest radiograph

Chest radiograph

PA and AP films

The standard view is posteroanterior (PA). The patient stands facing the film, clasping it in both arms, the X-rays pass through the chest *from posterior to anterior*, and the films are taken in full inspiration to maximize the size of the lung fields. This has advantages over the anteroposterior (AP) view, used for portable X-rays in immobile patients (where the plate is placed under the patient's back and the X-rays pass through the patient from *anterior to posterior*), namely:

- The scapulae are rotated away so they obscure the lung fields less.
- The heart shadow is smaller, so more of the lung fields are visible
- The relative sizes of the lung fields and the cardiac silhouette can be reliably used to comment on cardiomegaly and mediastinal widening.

Lateral chest films

Lateral films are used to further define abnormalities identified in the lung fields on the PA and AP films. A left lateral film is taken to define abnormalities in the left thorax. The patient's left side is placed against the film, which is labelled 'L'. Right lateral views are taken in a similar way (Fig. 1.1).

System for describing chest X-rays

- Confirm the name and age of the patient, and the date of the film.
- Define the film (AP, PA, or lateral, erect or supine, contrast or plain).
- Check that the film is oriented correctly!
- Check that it is correctly aligned (clavicle and ribs should be symmetrically aligned with sternum and spinous processes), shows all of the lung fields from apex to costophrenic angle, and has adequate exposure (you should see the vertebral bodies through the heart).
- Comment on presence and position of endotracheal (ET) or nasogastric (NG) tube, catheters (pulmonary artery (PA) or central line), clips, drains, prosthetic valves, rings, wires.
- Now examine the bony tissues looking for fractures and dislocations, lytic lesions, calcification, erosions, and extra ribs (the ribs are easier to see posteriorly than anteriorly).
- Next assess the soft tissue shadows of the neck, chest wall, breasts, and look at the diaphragms (right normally one rib higher than left).
- Examine the mediastinal shadow (Fig. 1.1: aortic knob 8; pulmonary trunk 9; LA appendage 10, LV 11; inferior vena cava (IVC) 12; right atrium (RA) 13; superior vena cava (SVC) 14).
- Then look at the airways, including the trachea, main bronchi, and lung fields (normal a deep breath in can inflate lungs to 10th rib posteriorly or 6th rib anteriorly), fissures, and vascularity of lung fields.
- Tips:
  - Trachea should be midline, opacities should be symmetrical.
  - It takes 400mL of fluid to blunt the costophrenic angles.
  - In the supine patient fluid level is replaced by diffuse opacity.
  - If one hemithorax is more opaque, either you have a collection on the opaque side, or a pneumothorax on the other side.

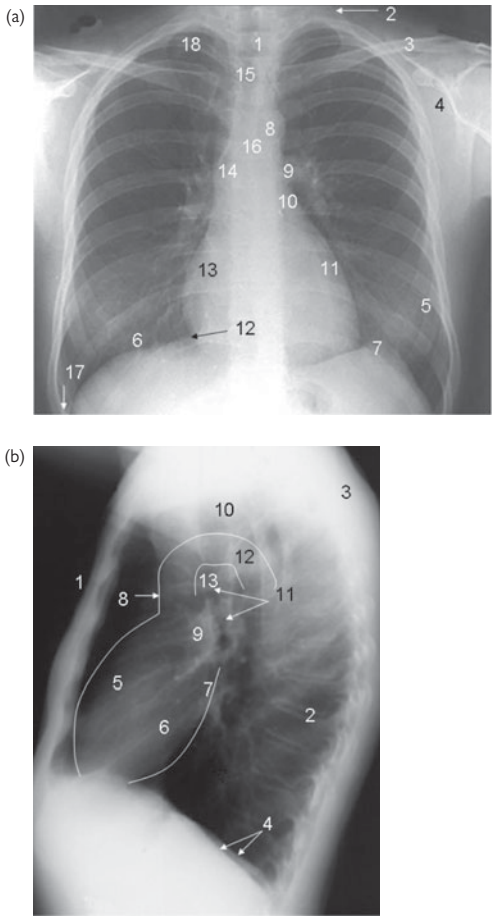


Fig. 1.1 (a) Normal CXR. 1, spinous process of T12; 2, 1st left rib; 3, left clavicle; 4, body of left scapula; 5, left 8th rib; 6, right hemidiaphragm; 7, left hemidiaphragm; 8, aortic knuckle; 9, pulmonary trunk; 10, left atrial appendage; 11, left ventricle; 12, inferior vena cava; 13, right atrium; 14, superior vena cava; 15, trachea; 16, carina; 17, right costophrenic angle; 18, apex right lung. (b) Plain lateral CXR. 1, manubriosternal junction; 2, body of thoracic vertebrae; 3, body of scapula; 4, right and left hemidiaphragms; 5, right ventricle; 6, left ventricle; 7, left atrium; 8, ascending aorta; 9, pulmonary trunk; 10, proximal trachea; 11, distal trachea; 12, distal arch of aorta; 13, aortopulmonary window.

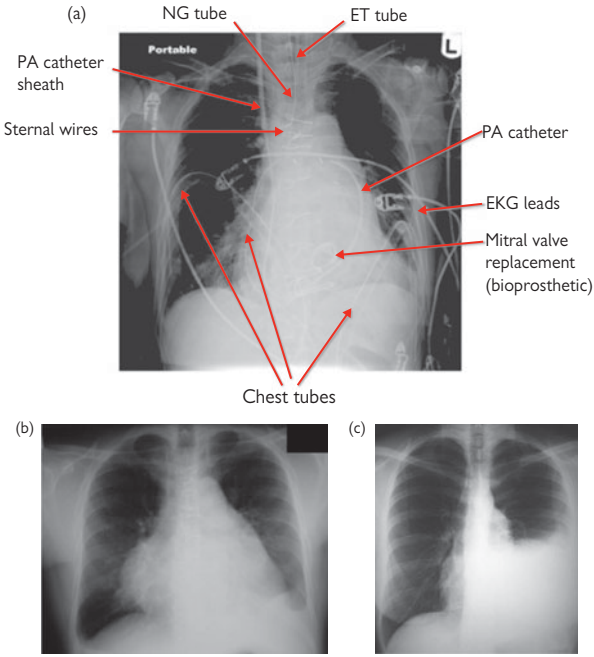


Fig. 1.2 Postoperative films. (a) Early film showing the position of most commonly seen lines and catheters. A portable X-ray is done on immediate arrival in the ICU. Check the position of the ET tube (should be 1 inch (2.5cm) above the carina), NG tube (should be visible in stomach), and PA catheter (usually in right pulmonary artery, check it is not coiled, in too far, or malpositioned, e.g. in IVC). Look for collections (remember these may just be diffuse haze in supine patients), pneumothorax, and lobar collapse. (b) Pericardial effusion—globular heart shadow, CT, and echo are diagnostic, and echo is required for diagnosis of tamponade. (c) Left pleural effusion in erect patient. Note concave upwards meniscus suggestive of fluid. If there is a flat fluid level, this suggests hydropneumothorax.

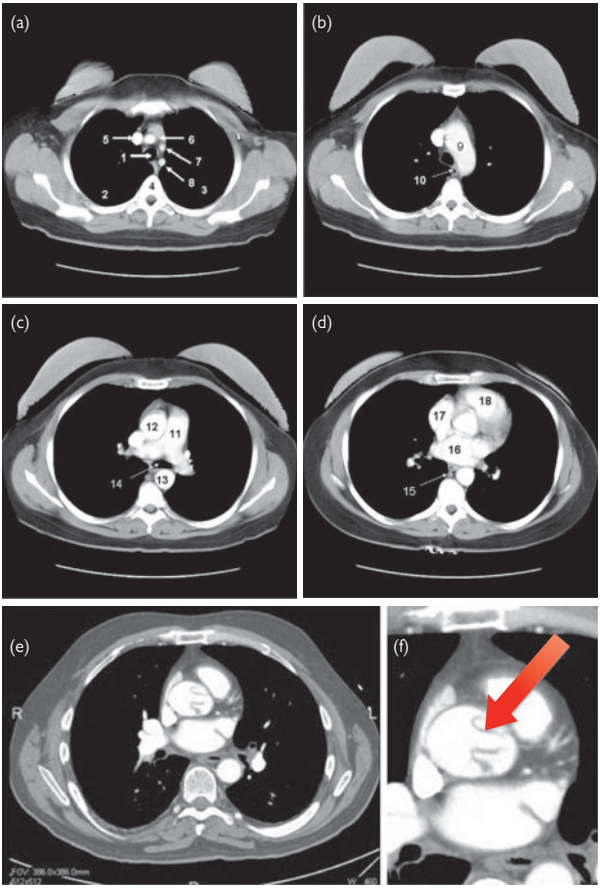


Fig. 1.3 (a–d) Normal contrast CT; 1, trachea; 2, right lung; 3, left lung; 4, body of T3; 5, SVC full of contrast; 6, brachiocephalic artery; 7, left common carotid; 8, left subclavian; 9, aortic arch; 10, esophagus; 11, pulmonary trunk; 12, ascending aorta; 13, descending aorta; 14, carina; 15, hemiazygous vein; 16, left atrium; 17, right atrium; 18, right ventricle. (e) CT with iv contrast showing type A dissection. (f) Detail (arrow indicates obvious dissection flap, which is often less prominent and more concave).

Computed tomography

Principles of computed tomography

CT uses the same radiation as standard X-rays, but moves the X-ray source round the patient. The attenuation of each point is measured on a scale of *Hounsfield units* (HU) which assigns water 0HU, and ranges from +1000 (white) to −1000 (black). Bone is +400 to +1000 (white), contrast is +800 (white), soft tissue is +40 to +80 (gray), lung is −500 (gray to black). As the eye can't distinguish 2000 shades of gray, a limited range of HU or *window width (WW)* is selected for display once images are acquired, around a central *window level (WL)*, depending on the tissue of interest. A ***mediastinal or soft tissue window*** has a WL of +40 HU (Fig. 1.4a), a ***lung window*** has a WL of –600 (Fig. 1.4b) showing detailed lung parenchyma, which cannot be seen on soft tissue windows. IV contrast is used to define vascular structures.

Types of CT scanners

- *Conventional CT scanners*: X-ray source and detectors are unable to rotate continuously. The table moves the patient a set distance, stopping between each slice so the next slice can be taken.
- *Spiral CT scanners*: the majority of conventional scanners have been replaced by spiral CT which incorporate 'slip rings' that remove the need for a physical connection between the X-ray tube and the power cables, meaning that the ring can rotate in one direction indefinitely. The table moves the patient continuously while the image is being acquired. Advantages of spiral CT over conventional include:
  - Faster, higher-resolution images allowing angiography—e.g. coronary and pulmonary angiography and standard vascular views.
  - Any plane and 3-dimensional (3D) images can be constructed.
- *Multislice CT scanners*: instead of the single row of detectors used in conventional and spiral scanners, multislice CT scanners have several rows of detectors. Tube rotation is also much faster. Advantages:
  - Image acquisition is several times faster than spiral CT scanners, so they are used in unstable and uncooperative patients, and vascular imaging, as scan time and movement artefact is minimized.
  - Resolution is significantly better allowing video footage, e.g. of valves.

CT scans in perioperative management of cardiac surgery patients

Chest CT is particularly helpful in cardiac surgical patients:

- Reoperations: identifying vascular structures, e.g. patent grafts, aorta, RV at particular risk during sternal re-entry (📖 p347).
- Valve patients at low risk of coronary disease may elect to have *CT coronary angiography* rather than cardiac catheterization to evaluate coronary disease preoperatively. CT angiography is very specific and sensitive for coronary disease. Any significant abnormality should be evaluated with standard coronary angiography (so little to gain from evaluating higher risk patients with this modality). Contrast load and radiation exposure are very similar for invasive and CT angiography.
- Diagnosis and evaluation of aortic aneurysm and dissection (Fig. 1.3e).
- Evaluation of mediastinitis, sternal dehiscence, PE.

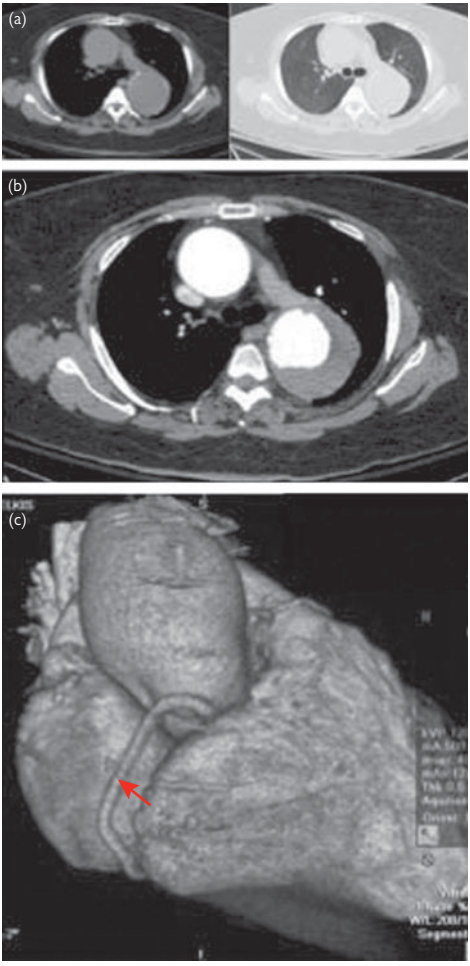


Fig.1.4 (a) Non-contrast CT of chest showing large ascending and descending aortic aneurysm (mediastinal windows on the left and lung windows on the right). Calcification is visible in the wall of the descending aorta but little other information about the vessel lumen can be derived. (b) The same patient and view after administration of iv contrast now clearly showing evidence of a type B dissection, with intraluminal thrombus in the descending aorta. (c) 3D reconstruction of aneurysmal ascending aorta showing right coronary artery (arrowed) in atrioventricular groove.

Cardiac catheterization

Key facts

- Cardiac catheterization may be right and/or left heart, diagnostic or therapeutic and allows:
- ***Left heart catheterization*** (via the femoral or radial artery):
  - Measurement of LA, LV, and aortic pressures and transmitral and transaortic pressure gradients.
  - Calculation of mitral and aortic valve areas.
  - ***Coronary angiography*** (the commonest indication) and ventriculography which requires substantially more contrast.
  - PCI, percutaneous balloon mitral valvuloplasty (BMV), transcatheter aortic valve implantation (TAVI), LA appendage occlusion.
- ***Right heart catheterization*** (via the femoral, internal jugular, or subclavian vein) is essentially the same as floating a Swann–Ganz catheter:
  - Measurement of RA, RV, and PA pressures, pulmonary artery wedge pressure (PAWP) and saturations including SvO_2 (mixed venous oxygen saturation).
  - Calculation of peripheral vascular resistance (PVR), CO, shunts.
  - Pulmonary angiography.
  - Transseptal puncture for LA access.

Contraindications

These are few. The patient may be too unwell (acute type A dissection) to tolerate the procedure. Patients with AS or mobile vegetations should not have ventriculography or pull back gradients performed as crossing the aortic valve carries a risk of stroke, and hemodynamic data can be obtained more safely by echocardiography. Similarly, patients with renal dysfunction should not have ventriculograms as the increased contrast load will exacerbate renal impairment significantly.

Principles of coronary angiography

Technique

The patient is usually awake but sedated and supine. Access is via a sheath using a modified Seldinger technique. Radio-opaque contrast is injected into the coronary ostia in turn via a size 4 Judkins catheter (2–3mm diameter). The X-ray tube and camera are opposite each other on a C-arm, and can be rotated cranially and caudally, and from left to right to achieve a range of views of the coronary anatomy. Biplane imaging uses two cameras to obtain simultaneous images from two different angles, reducing radiation exposure. Moving images are captured on digital CDs which have superseded cine film. Patients may be given 300mg of clopidogrel or 75mg of aspirin pre- and post-PCI.

Complications

- Complications requiring emergency bypass surgery (left anterior descending coronary artery (LAD) dissection, acute LAD occlusion, cardiac tamponade) 1:1000 catheterizations.
- Major adverse events (e.g. stroke) 1%, death 0.01%.
- Failure to occlude the puncture site after removal can result in extensive hematoma, which may be retroperitoneal if femoral puncture high.

Coronary anatomy is described on 📖 pp20–21 and pp46–47. Interpretation of coronary angiograms is described below.

Interpretation of coronary angiography

- Look at each lesion in several views and the whole film.
- Eccentric plaques will look most significant when viewed end on, and may almost disappear when viewed perpendicularly.
- Comment on the vessels systemically.
- Lesions are compared to adjacent 'normal' segments: in diffusely diseased vessels this can lead to underestimating severity.
- A reduction in ***vessel diameter*** of 50% corresponds to a 75% reduction in ***cross-sectional area***, and a 70% reduction in vessel diameter corresponds to a reduction of 90% in cross-sectional area (Fig. 1.5).
- A stenosis is ***significant*** if it is >50% the lumen diameter as it reduces the normal 3–4 x flow reserve, >70% lumen diameter stenosis eliminates any capacity to increase above resting flow, and >90% stenosis usually means TIMI grade 1 or 2 flow (📖 p22).
- Remember occluded vessels frequently fill late, retrogradely from shots of other vessels: most commonly the PDA and right coronary artery (RCA) from the LAD, and sometimes the LAD from the PDA.
- Beware mistaking a large diagonal for a completely occluded LAD: always look at the left anterior oblique (LAO) view before commenting on the LAD.
  - Start with a summary after viewing ***all*** views: 'This is a 75-year-old male with 3-vessel disease and moderate LV…'.
  - Describe the LMS, best seen in the right anterior oblique (RAO) view and the LAO caudal.
  - Describe the circumflex system, best seen in the RAO view.
  - Describe the LAD and diagonals, best seen in the LAO cranial.
  - Describe the RCA and PDA, and the dominance.
- Describe the ventricular function, any regional akinesia, MR, and calcification of the coronary vessels or valves.

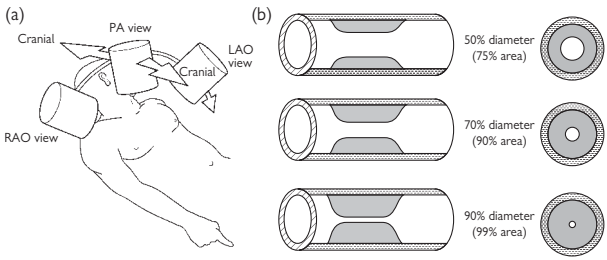


Fig. 1.5 (a) The orientation of the X-ray box and camera for the various views. (b) Relationship between angiographic and actual lumen narrowing.

Interpreting coronary imaging

Key facts

- Recognizing views and vessels rapidly becomes pattern recognition, but following a systematic approach will help avoid common mistakes.
- Identify each view, look at all the views, look at each vessel in multiple views, look at the whole clip for each view (some vessels only fill late).

Which view is it, which vessel is it?

- In a RAO view ribs go down to the right of the image, which is on the right of the spine (catheter in aorta).
- In a LAO view the ribs go down to the left of the image which is on the left of the spine (catheter in aorta).
- In cranial views the diaphragm is superimposed over most of the picture, whereas in caudal views it is hardly seen.
- The circumflex is the vessel closest to the spine (or the catheter in the descending aorta) in every image.
- The LAD is the vessel that travels towards the apex, it is usually the longest vessel, and it has septals coming off it at 90°.

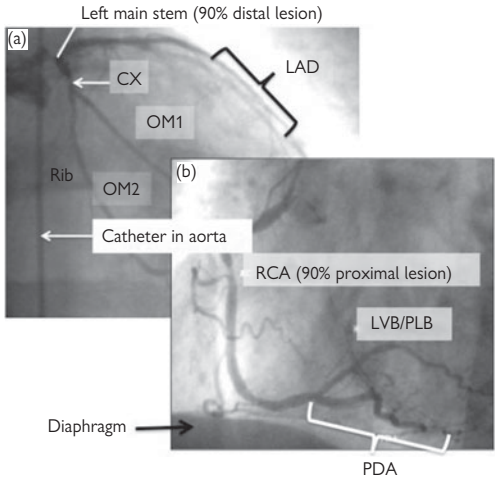


Fig. 1.6 Examples of coronary angiograms. (a) RAO view of left coronary tree showing LMS disease. The circumflex is always the most posterior vessel, i.e., it is closest to the spine, and the descending aorta in which the catheter can usually be seen. (b) RAO view of right coronary tree showing proximal RCA stenosis. You can just about make out the short septals coming off the PDA at 90°. These help distinguish the PDA from the LVB.

Coronary anatomy (see also 📖 pp46–47)

- Branches of the ***right coronary artery*** in order:
  - 1) Conus branch, 2) SA nodal branch, 3) RV branch, 4) acute marginal, 5) AV nodal artery (60%), 6) in right-dominant patients (85%) the RCA then bifurcates into PDA which gives off inferior septals at 90°, and 7) LVB or posterolateral branches.
  - Proximal RCA is up to the RV branch, mid up to the acute marginal, and distal up to the bifurcation.
- Branches of the ***left coronary artery:***
  - 1) LMS which bifurcates (occasionally trifurcates into ramus intermediate branch) into 2) LAD which gives off septals at 90° and diagonals, and 3) Cx which gives off OMs and may give rise to AV nodal artery (40%) and in left-dominant patients PDA (10%) and LVB or posterolateral branches.
  - The proximal LAD is up to the 1st septal, mid LAD until the 2nd diagonal, and distal LAD after the 2nd diagonal.

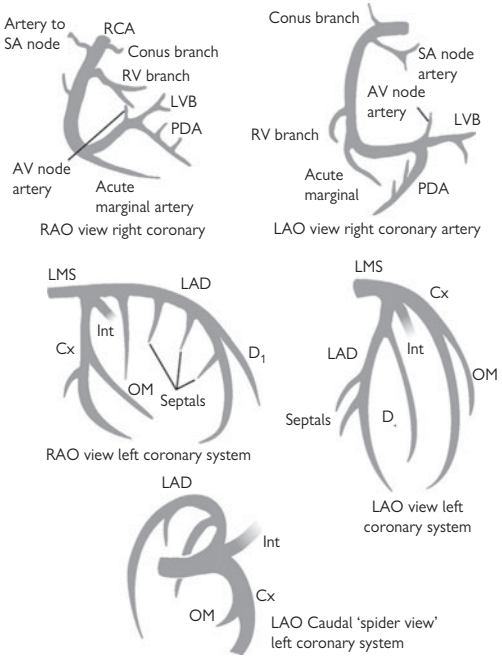


Fig. 1.7 The five common angiographic views. RCA, right coronary artery; RV, right ventricular; SA, sinoatrial; AV, atrioventricular; LVB, lateral ventricular branch; PDA, posterior descending artery; LMS, left main stem; LAD, left anterior descending; D, diagonal; Int, intermediate; Cx, circumflex; OM, obtuse marginal.

Cardiac hemodynamics

Key facts

- Severity of coronary stenosis may be misrepresented by angiography: additional information can be obtained by pressure measurements, e.g., fractional flow reserve (FFR), and intravascular ultrasound (IVUS).
- When echocardiography is equivocal, cardiac catheterization can provide useful additional information on severity of valve lesions, pulmonary hypertension, ventricular function, and intracardiac shunts.

Coronary hemodynamics

Fractional flow reserve

This is essentially the ratio between the BP proximal to and distal to a lesion, when flow is maximal. A FFR <75% (i.e., <75% of maximal flow in a non-diseased coronary) indicates a significant lesion and there is some data to suggest it is safe to watch (rather than intervene) on patients with higher FFRs. The test is performed at maximal blood flow which is induced by giving adenosine or papaverine (and which is 4–5 x resting blood flow in the normal subject), hence it assesses coronary reserve.

TIMI flow

Thrombolysis in myocardial infarction (TIMI) scale grades coronary flow:

- TIMI 0 (no perfusion): absence of antegrade flow beyond a lesion.
- TIMI 1 (penetration no perfusion): incomplete filling distal coronary bed.
- TIMI 2 (partial reperfusion): delayed but complete filling distally.
- TIMI 3 (complete perfusion): normal flow.

Assessment of valve disease

Quantifying valve stenosis by cardiac catheterization

- Transaortic valve pressure gradient is estimated by measuring LV pressure and either simultaneously measuring aortic pressure distal to the valve (*mean and peak to peak gradients*) or by '*pull-back*' where the catheter is then pulled back into the aorta to record pressure. The most physiologic value is the mean gradient (Fig. 1.8).
- Mitral pressure gradients are measured by simultaneous recording of PAWP or LA pressure (LAP) obtained by transseptal puncture, and LV end-diastolic pressure (LVEDP).
- *Valve area* is estimated using the *Gorlin equation* (essentially valve area = flow/velocity) or *area = cardiac output /√pressure gradient.* Accurate estimation of cardiac output is particularly critical, and particularly open to error in low cardiac output states (📖 p492) when dobutamine stress testing can be used as a diagnostic adjunct (📖 p25).

Grossman grading of regurgitation

- Grade I: faint opacification of proximal chamber, clears with one beat.
- Grade II: opacification of proximal less than distal chamber.
- Grade III: opacification proximal chamber equals distal chamber.
- Grade IV: complete dense opacification of proximal chamber with a constant density greater than in the distal chamber.

Calculation of cardiac output

There are multiple methods of calculating cardiac output during cardiac catheterization, all with potential errors.

- *Thermodilution* technique (📖 p135) involves injection of a bolus of normal saline into the proximal port, measurement of the pressure gradient between this and a distal port, which is entered into an equation incorporating the specific gravity and temperatures of the fluid and the patient's blood. TR, incorrect injection technique, and low cardiac output are all potential sources of error.
- The *Fick principle* is that the uptake (or release) of a substance (usually VO_2) by a substance is equal to the product of the arteriovenous concentration gradient and the cardiac output. In most labs VO_2 is not measured, and assumed to be 125mL/m^2. This method is more accurate than thermodilution in low output states, but should not be used in MR or AI, or patients receiving supplemental O_2.
- *Angiographic methods* involve tracing the LV end-diastolic and end-systolic images and converting these to volume using modelling.

Assessment of other pathology

- PVR = (mPAP − mLAP)/CO, where PAWP is substituted for LA pressure in the absence of pulmonary disease. Normal PVR is 3 Woods units. Fixed pulmonary hypertension cannot be reversed with NTG, inhaled NO or Flolan, or exercise (📖 p504).
- Cardiac catheterization is also used to differentiate between restrictive cardiomyopathy and constrictive pericarditis (📖 pp83, 488).
- Shunts can be detected and quantified using sampling of blood oximetry from IVC, SVC, multiple RA locations, PA and coronary sinus and artery: Qp:Qs = (SaO_2 − MVO_2/PVO_2 − PAO_2).

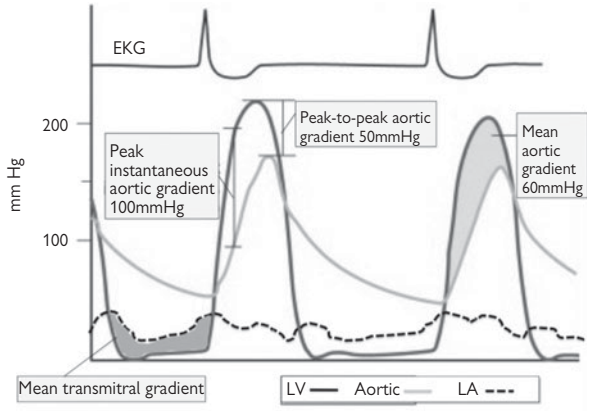


Fig. 1.8 Quantifying valve stenosis by cardiac catheterization.

Echocardiography

Key facts

Echocardiography is ultrasound used to assess cardiac conditions. There are two main approaches: transthoracic (TTE) and transesophageal echocardiography (TEE), and four modalities are used commonly (Box 1.2).

Box 1.2 Terminology in echocardiography

- *B-mode* (brightness mode), also called 2D or real-time, provides a 2D grayscale picture of a single plane across the heart. The intensity of the echo is related to the brightness of signal.
- *M-mode* (motion mode) is more accurate than B-mode, so it is used for measurements, but it only yields a 1D image of structures plotted on the y-axis against time on the x-axis.
- *D-mode* (Doppler) uses Doppler frequency shifts to measure flow velocities and hence calculate gradients and valve areas. Only moving targets are displayed. Duplex US combines Doppler and grayscale images to correlate flow with fixed structures.
- *Colour flow imaging* is a colour-enhanced duplex Doppler echo which combines M-mode and D-mode with blood flow imaging. Most systems code blood flow **a**way from the transducer as **b**lue, and blood flow **t**owards the transducer as **r**ed (BART).
- *3D echo* allows 3D visualization of the cardiac structures. It is particularly useful when viewing the morphology of the mitral valve.

Principles of echocardiography

US is a waveform of 2–5MHz that passes through tissues, with reflection from tissue interfaces. US waves are both generated and detected by piezoelectric crystals which act as transducers, converting electrical to mechanical energy and vice versa. Reflected waves are calibrated to form an image. The Doppler effect describes how reflected waves from a moving target change in frequency, increasing if the target is moving towards the wave source, and decreasing if the target is moving away. The frequency change, or *Doppler shift*, is proportional to the speed of the object, which can be quantified as an audible signal, or color (see Box 1.2). The frequencies used cannot penetrate air or bone, which explains the need for certain views to assess some structures.

Echo quantification of valve stenosis

- Bernouilli equation: transvalvular gradient (mmHg) = $4v_m^2$
- Continuity equation: valve area (cm^2) = SV/(ET x 85 x v_m)
- In AS: mean transvalvular gradient = $2.4v_{max}^2$

v_m= mean velocity of blood (m/s), v_{max} = maximum velocity of blood, VF= valve flow, VG = transvalvular gradient, SV = stroke volume, ET= ejection time.

Normal cardiac dimensions

Echocardiography allows precise measurement of cardiac dimensions. Septal and posterior wall thickness in systole and diastole allow calculation of fractional shortening, which is a crude index of LV systolic function (which is expressed as a percentage but should not be confused with ejection fraction), and LV mass. See Table 1.3.

Table 1.3 Normal adult ranges in M-mode echocardiography

Structure	Parameter	Range
Left ventricle	End-systolic diameter	25–41mm
	End-diastolic diameter	35–56mm
	Shortening fraction	30–40%
	Ejection fraction	50–65%
	Mass	60–124g/m^2
Aortic root	Diameter	20–37mm
	Cusp separation	15–26mm
Left atrium		19–40mm
Right ventricle		7–23mm

Assessing LV function

LV function can be assessed quantitatively and qualitatively. Qualitative assessments involve scoring regional wall motion and categorizing LV function as normal, mild, moderate, or severe impairment. Quantitative methods, e.g. fractional shortening and calculations of ejection fraction are still subject to operator variability. The nine-segment model for regional wall motion assessment is shown in Fig. 1.9.

Stress echocardiography

Stress echocardiography stresses the myocardium either chemically or via exercise protocols to gain information about myocardial viability and regional perfusion. Standard treadmill protocols such as the Bruce protocol are used for exercise testing: they give a limited time in which to image the heart and hyperventilation may make imaging difficult. The main drug used in pharmacological stress echocardiography is dobutamine in incremental low- or high-dose protocols.

- Net outward systolic motion and failure of a segment to thicken during systole suggests that the segment is infarcted.
- Reduction of wall motion during stress, in a segment that contracts when the patient is at rest, suggests flow-limiting coronary stenoses.
- During low-dose dobutamine protocols, improvement of function in a segment that is impaired at rest suggests that the segment contains stunned or hibernating myocardium (📖 pp74–75).
- Increase in a moderate gradient across a critically stenosed aortic valve suggests that myocardium will respond well to valve replacement.
- Papillary muscle ischemia may cause MR as a result of an inducible septal or posterior wall motion defect.

Echo interpretation

TEE versus TTE

- TEE has finer resolution than TTE (down to 2–3mm).
- TEE is better than TTE at imaging the posterior heart (e.g., identifying LA appendage thrombus), and the ascending aorta.
- TEE can be carried out under light sedation. Relative contraindications include esophageal strictures (use pediatric probe), pouches and varices, respiratory compromise, and cervical pathology.

Commonly used views

TEE can be distinguished from TTE as: (1) films are usually labelled, (2) in a TEE the left atrium is the chamber closest to the transducer and hence at the apex of the image triangle, and (3) the AV valves are concave downwards in most views. Standard TEE views are shown in Fig. 1.9. Tips for orientating yourself when looking at TEE films:

- The apex of the TEE image is the probe in the esophagus, i.e., posterior and the left atrium is almost always at the apex of the triangle (Fig. 1.9).
- Right ventricle and atrium are the most anterior structures.
- The aortic valve is adjacent to the anterior leaflet of the mitral valve.
- The inter-atrial septum points at the non-coronary cusp of the aortic valve in the short-axis view (see Fig. 1.9).
- In the long-axis view the aortic valve may be on the left or right of the image depending on the probe orientation, but it is always adjacent to the interventricular septum.

Common sources of error

Image quality in TTE is highly dependent on patient build (obese, emphysematous, postoperative patients are difficult to image), patient position, probe position and contact, and machine settings. Other errors include:

- Failure to align the US beam parallel to the flow of blood thus underestimating its velocity and hence transvalvular gradient.
- Aligning the beam obliquely across a chamber and overestimating size.
- Measuring eccentric jets, or other signals (the jet of MR may be mistaken for AS if views are poor) instead of the main blood flow.

Echocardiographic grading of regurgitation

There are a number of qualitative and quantitative variables each of which can be used to grade regurgitation. This box summarizes some of the more commonly used indices.

Aortic regurgitation

- Mild: central jet width <25% of LVOT, regurgitant fraction <30%.
- Moderate: values intermediate to mild and severe.
- Severe: central jet width >65% of LVOT, regurgitant fraction >50%.

Mitral regurgitation

- Mild: central jet <20% of LA area, regurgitant fraction <30%.
- Moderate: values intermediate between mild and severe.
- Severe; central jet >40% of LA area, regurgitant fraction >50%, jet impinging on wall of LA, systolic flow reversal in pulmonary veins, effective regurgitant orifice (ERO) >40mm.

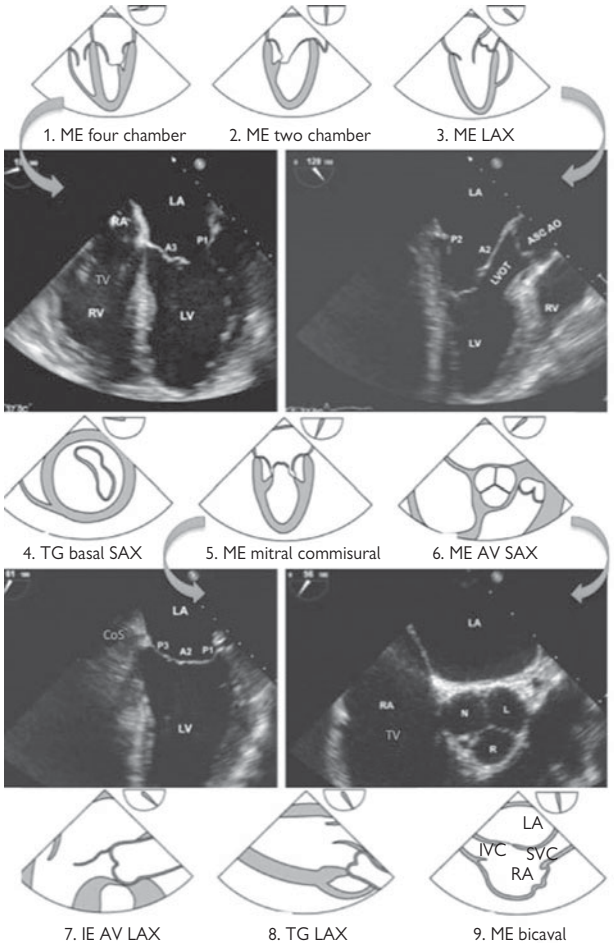


Fig. 1.9 9 of 20 standard TOE views. ME, mid-oesophageal; LAX, long axis; SAX, short axis; TG, transgastric; LV, left ventricle; RV, right ventricle; RA, right atrium; LA, left atrium; CoS, coronary sinus; TV tricuspid valve; N, non-coronary cusp; L, left coronary cusp; R, right coronary cusp; LVOT, left ventricular outflow tract. A1, A2, P1, P3, anterior and posterior mitral leaflets (see 🕮 p1 for terminology).

Perfusion and viability

Key facts

- The historic view that chronic ventricular dysfunction was synonymous with infarction was challenged in the 1970s by the observation that it could be reversed by revascularization.
- The term '***hibernating myocardium***' was coined to describe the theory that chronic LV dysfunction was an adaptive response to chronically reduced coronary blood flow at rest.
- More recent studies suggest that hibernation is really due to repetitive *stunning* (📖 pp74–75): myocardium is adequately perfused at rest, but on exercise becomes ischemic because of impaired coronary flow reserve.
- Characteristics of hibernating myocardial include:
  - Reduced contractility (hypokinesis rather than akinesis).
  - Reduced perfusion (usually evident on exercise rather than rest).
  - Viable (i.e., not infarcted) myocardium (evidenced by hypokinesis).
  - Functional improvement with revascularization, which can sometimes be predicted by similar improvement with intropes or exercise (***contractile reserve***).
- Detecting hibernating or viable myocardium is clinically important in patients with poor LV function and predominantly dyspneic symptoms (rather than angina) being considered for coronary artery bypass graft (CABG): the absence of viability suggests much lower prognostic and symptomatic benefit will be obtained from high-risk surgery (📖 p299).
- Key modalities for assessment of viability include nuclear imaging, stress echocardiography, and magnetic resonance imaging (MRI).

Nuclear imaging

Nuclear imaging of the heart allows ventriculography and perfusion studies. Isotopes may either be γ-ray or positron-emitting radionuclides:

Conventional radionuclide imaging

^{99m}Technetium is used to make compounds such as sestamibi and tetrofosmin (Table 1.4), which are replacing cheaper ^{201}thallium-labeled compounds. These radiolabeled compounds are taken up by red blood cells (RBCs). Decay produces γ-rays which hit a sodium iodide crystal causing scintillations detected by a photomultiplier, providing images of regional perfusion.

Single-positron emission computed tomography (SPECT)

Two gamma cameras and computerized reconstruction allow images from technetium or thallium tests to be reconstructed in any plane. Stress imaging (after exercise, dobutamine, dipyridamole, or adenosine) is performed first: if this is normal, subsequent rest imaging does not need to be performed. If it is abnormal, rest images are obtained: fixed perfusion defects represent infarcts, reversible defects suggest hibernation.

Positron emission tomography (PET)

^{18}Fluoro-2-deoxyglucose (^{18}FDG) is most widely used: it is taken up by myocytes, reflecting glucose metabolism during ischemia. PET scans offer much better spatial resolution than the 1cm offered by SPECT but routine use is limited by very high costs.

Table 1.4 Commonly used radiopharmaceuticals

Radionuclide	Compound	Imaging
^{99m}Tc	^{99m}Tc-labeled RBCs ^{99m}Tc MIBI ^{99m}Tc pyrophosphate	Blood pool studies Ventriculography Myocardial perfusion
^{201}Tl	^{201}TlCl	Hibernation
^{18}FDG ^{13}N		Hibernation Myocardial perfusion
^{13}O		Oxygen uptake

Interpreting images

Ventriculography

Nuclear imaging can be used to provide an accurate, relatively non-invasive, reproducible ejection fraction and intracardiac shunts.

Myocardial perfusion imaging

Regional blood flow is imaged after stress then at rest. Stressors include adenosine, dopamine, and exercise (see 📖 p25). During stress, territories supplied by a vessel with a flow-limiting stenosis receive relatively less blood than adjoining normal territories. A defect in the post-stress scan which normalizes after rest (reversible defect) usually represents reversible ischemia, which may be improved by revascularization (Fig. 1.10). A photon-deficient area which remains unchanged in both phases of the study is called a fixed defect, and usually represents an area of infarction, which will not be improved by revascularization.

Magnetic resonance imaging

Cardiac magnetic resonance (CMR) imaging can assess myocardial viability through measurement of end-diastolic wall thickness, dobutamine stress (which identifies contractile reserve), and contrast enhancement revealing scar tissue, as well as accurate measurement of LV and RV volumes and ejection fraction. Late enhancement can also be demonstrated on CT along with measurement of wall thickness and volumes.

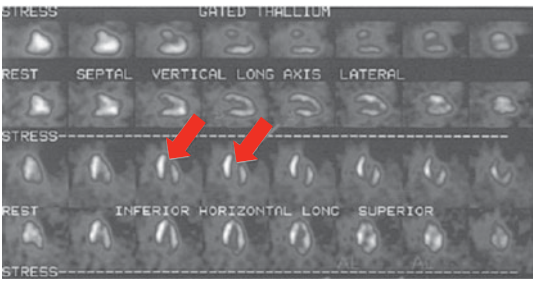


Fig. 1.10 Thallium scan showing apical (A) perfusion defects after stress (arrowed) which reverse on rest images below, suggesting reversible ischaemia.

General pre-op checklist

Box 1.3 contains an overview of planning an elective operating list in cardiac surgery. For emergency surgery, make sure that the following are available: an ICU bed; a surgeon; anesthesiology team; an operating room (OR); two scrub nurses; a perfusionist; blood.

Box 1.3 Elective list checklist

- Always see the patients you are operating on: always take your own history and do your own examination.
- Consent the patient (📖 p257).
- Review the bloods and check that they have been cross-matched.
- Review the CXR and make sure all relevant films are available.
- Review echo/angiography, repeat studies >6 months old.
- Does the patient need any additional investigations?
- Decide on the order of the list with the surgeon:
  - Sick, complex patients who medically will not tolerate cancellation, and who would benefit from admission to ICU early in the day when there is full availability of imaging and support services, are best placed first on the list.
  - Patients with infection risks are placed last on the list.
  - The availability of equipment may dictate the running order.
- Speak to the anesthesiologist, particularly if there are last-minute changes.
- Make sure the nursing staff know the list order: a copy of the list should be on every location with a patient on the list, ICU, and OR.
- When writing the list, in addition to operation, specify radial artery and side, off-pump, redo, femoral bypass, thoracotomy, and side.
- Homografts (📖 p386) may need to be ordered in advance.
- Pacing technicians should be contacted for patients with permanent pacemaker (PPMs).
- Theatres should be notified of patients with latex allergies several days in advance if possible.
- If the patient is an outlier on a ward not used to preparing patients for surgery, make sure that the nursing staff know that the patient must be starved from midnight, be dressed in a gown, on a stretcher, and shaved at least 30 minutes before the send time.
- Let ICU know which patients are appropriate for fast-tracking.
- Tell the perfusionist if the patient is anemic (they may decide to prime the pump with blood and you will need to XM extra), weighs <60kg (they may want to minimize prime volume to avoid hemodilution), and if the patient has a high creatinine (the perfusionist can filter the patient on bypass).

Predicting mortality risk

Several *risk stratification* tools estimate operative risk. Most institutions submit mandatory data to registries which may publish institution and *surgeon-specific risk-adjusted comparative outcome data*, i.e., mortality and complication rates which take into account certain patient comorbidities,

demographic factors, and nature of surgery. Data is usually published in the format where raw mortality data for *index operations*, e.g., isolated CABG, is provided for individual institutions or surgeons, highlighting any that fall outside 2 standard deviations above or below the risk-adjusted mean. This may have contributed to lowering mortality rates, possibly through ***risk-averse behaviour*** leading to bias against high-risk patients. The main risk stratification systems used are:

- *EuroSCORE* (see inside front cover): based on European CABG patients, available as simple additive and as a more accurate logistic version, general tendency to overestimate operative risk.
- *STS* (Society of Thoracic Surgeons): similar model validated in North American patient groups, freely available as online calculator.
- *Parsonnet score*: one of the earliest systems, not widely used now.

Summary of preoperative management

Warfarin: stop 3–5 days preoperatively. In high-risk patients (e.g., mechanical valve) admit early and commence patient on heparin, which should be stopped 6 hours before surgery.

Aspirin and clopidogrel: aspirin should be stopped 5 days pre-op, clopidogrel 10 days pre-op in low-risk patients (see 📖 p4).

ACE inhibitors: withhold on the day before surgery.

Diabetes: non-insulin-dependent diabetics should have normal medication, but withhold long-acting hypoglycaemics (gliclazide) on the morning of surgery. Insulin-dependent diabetics should be first on the list, and have their morning insulin replaced by a sliding scale and iv fluids.

Carotid bruits: obtain a carotid Doppler if the patient has a history of cerebrovascular accident (CVA), TIA, or severe peripheral vascular disease. Carotid intervention may be indicated for internal carotid stenoses >70% especially if the contralateral side is tightly stenosed, to reduce the risk of CVA (📖 p279). Discuss with vascular surgeon. Maintain high perfusion pressures.

Murmurs: all patients should have recent pre-op TTE: review it!

Anemia: unexplained anemia requires full outpatient investigation. Surgeon may elect to prime the pump with blood rather than crystalloid. Patients are not normally transfused preoperatively.

Neutrophilia: valve patients should be cancelled and a source of infection found and treated. CABG may go ahead if the WCC <12.0 and the CRP is normal. Always discuss with the surgical team.

Respiratory impairment: admit high-risk patients for physiotherapy, nebulizers, and education.

Renal impairment: acute deterioration should be treated before surgery: stop nephrotoxic drugs including ACE inhibitors. Consider iv fluids.

MRSA: this should be treated preoperatively: consult the infection control team to ensure the appropriate treatment.

Lung lesions: these should be investigated preoperatively with a lateral film and CT. Isolated lung lesions of non-infective origin are not usually a contraindication to coronary artery surgery.

This page intentionally left blank